



INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI

(Established by Ministry of Human Resource Development, Govt. of India)

Sector 81, S A S Nagar, Manauli PO Punjab 140306

MID-SEMESTER EXAM
INTRODUCTION TO ENVIRONMENTAL SCIENCE
Please read the questions carefully before answering.

EES202 Date: 26.10.2024

Total Marks: 20

Section A: Choose the correct option (1 x 5 Marks; 1 for correct answer and -1 for wrong answer)

1) For a locally unstable atmosphere, where T_{atm} and T_A are atmospheric and adiabatic lapse rates respectively, which of the following is true:

- a) $T = T_A$ (b) $T > T_A$ (c) $T = -T_A$ (d) $T < T_A$

2) Hadley circulation cell predicts transport of moisture from :

- (a) ITCZ to poles (b) poles to ITCZ (c) 60 N to 90 N (d) 60 S to 90 S

3) Atmospheric lapse rate is greater than zero in:

- (a) troposphere (b) stratosphere (c) mesosphere (d) asthenosphere

4) The number density of 365 ppm carbon dioxide at sea level for $P = 1013 \text{ hPa}$, $T = 0^\circ \text{C}$ is:

- (a) $9.8 \times 10^{21} \text{ molecules m}^{-3}$ (b) $4.5 \times 10^{21} \text{ molecules m}^{-3}$ (c) $9.6 \times 10^{19} \text{ molecules m}^{-3}$ (d) $2.69 \times 10^{25} \text{ molecules m}^{-3}$

5) 1 mole of moist air is:

- (a) heavier than 1 mole of dry air (b) weighs as much as 1 mole of dry air (c) always lighter than 1 mole of dry air (d) none of the above

6) The Solar constant for Venus is 2640 W m^{-2} and the planetary albedo of Venus is 0.75.

Calculate the effective temperature on Venus as observed from space assuming it does not have an atmosphere? (Stefan Boltzmann constant = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Even though Venus is closer to the Sun than the Earth, why is its effective surface temperature lower than that of the Earth in this sort of calculation ($T_{\text{eff for Earth}} = 255 \text{ K}$)? (2 + 1 = 3 Marks)

7) Explain using diagrams why good weather at surface is typically associated with high surface pressure whereas bad weather at surface is associated with low surface pressure. Why are most of the large deserts located at around 30 N latitude? (3 + 1 = 4 Marks)

8) Derive and show that the dry atmospheric adiabatic lapse rate in the Earth's atmosphere is 9.8 K/km . (4 Marks)

9) What does the ITCZ stand for and what is its importance for global atmospheric transport of air? (2 Marks)

10) Calculate the increase in gC per year for a gas X which increases at the rate of 4 ppmV of gas X /yr on an Earth like planet (1.2×10^{20} moles of air in its atmosphere) . Gas X has 4 carbon atoms and a molecular weight of 56 g mol^{-1} (2 Marks)